MIT VISHWAPRAYAG UNIVERSITY
School of Computing
End Semester Assessment SET D

Program Name: BCA SEM : III A Y : 2025-26 Date:
Course Code : AS2003 Course : Probability
Max Marks: 100 Duration: 180min

Instruction: 1. Answer ANY TWO full questions from SECTION I & II
2. Answer ANY ONE full question from SECTION III

SECTION I					
QN		Question	Marks	Level	CO
1	A				
	1)	Select the correct option. 1. Two events A and B are mutually exclusive. If P(A) = 0.4 and P(B) = 0.3, then the probability of the union, P(A \cup B), is: a) 0.12 b) 0.60 c) 0.70 d) 0.58 2. A fair coin is tossed three times. What is the probability of getting three heads (HHH)? a) 1/3 b) 1/2 c) 1/6 d) 1/8 3. Four fair coins are tossed once. What is the probability of getting exactly three heads? a) 1/4 b) 3/16 c) 1/8 d) 4/16 4. The set of all possible outcomes of a random experiment is called the: a) Event b) Probability c) Sample Point d) Sample Space 5. Two events A and B are considered independent if: a) $P(A \cap B) = 0$ b) $P(A \cap B) = P(A) + P(B)$ c) $P(A\|B) = P(A)$ d) $P(A \cup B) = 1$	5	L2	CO1
	2)	State True or False 1. If three events A, B, C are pairwise independent, they must be mutually independent. 2. The odds against an event with probability 1/4 are 3 : 1. 3. $P(A \cup B) = P(A) + P(B) – P(A \cap B)$ holds even if A and B are not independent.	3	L2	CO1
	B				
	1)	The probability of rain on a day is 0.4, and the probability of wind is 0.3, with both occurring together at 0.1. Use the addition rule to determine the probability of rain or wind or both.	3	L3	CO1
	2)	A witness identifies a suspect with 80% accuracy in good lighting and 50% in poor lighting. Suppose 70% of crimes occur in good lighting. If the witness identifies the suspect, calculate using Bayes' theorem the probability that the lighting was good.	3	L3	CO1

	3)	In a medical test, 2% of the population has a disease. The test is positive for 95% of those with the disease and 3% of those without. If a person tests positive, determine the conditional probability they have the disease.	3	L3	CO2
	4)	A pack of 52 cards is shuffled, and 4 cards are dealt. Calculate the probability that all 4 cards are from different suits.	3	L3	CO2
2	A				
	1)	Classify the following random variables as Discrete or Continuous random variables. 1.The voltage level of a battery at a random point in time. 2.The number of goals scored by a football team in a season. 3. The temperature inside a classroom at a randomly chosen moment.	3	L2	CO3
	2)	Select the correct option. 1. If p(x) is the pmf of a discrete random variable X, then P(X = 5) is equal to: a) p(5) b) ∫p(x)dx c) p(5) + p(6) d) None of these 2.For geometric distribution with p = 0.3, the probability that the first success occurs on the 4th trial is: a) $0.3 \times (0.7)^3$ b) $(0.7)^3 \times 0.3$ c) $(0.3)^4$ d) 0.7×0.3^3 3. In a binomial distribution B(n, p), the maximum possible value of X is: a) n b) n–1 c) np d) Infinite 4.Variance of a Bernoulli random variable with parameter p is: a) p b) p(1–p) c) p^2 d) 1–p 5.The pmf of a discrete uniform random variable taking values 1 to 6 is: a) P(X = k) = 1/6 for k = 1 to 6 b) P(X = k) = 1/5 for k = 1 to 6 c) P(X = k) = k/21 for k = 1 to 6 d) P(X = k) = 6/k for k = 1 to 6	5	L2	CO3
	B				
	1)	A lottery ticket number is a random integer from 000 to 999 (all three-digit combinations equally likely, including leading zeros). Find the probability that the sum of the three digits is exactly 15.	3	L3	CO1
	2)	The probability that a shooter hits a target is 0.6. He fires 7 shots independently. Find the probability that he hits the target exactly 5 times.	3	L3	CO1
	3)	A basketball player makes a free throw with probability 0.8. He keeps shooting until he makes the first basket. Find the probability that he needs more than 3 attempts.	3	L3	CO1
	4)	The pmf of X is p(x) = c/x for x = 2,3,4,5,6. Find the constant c and calculate E(X).	3	L3	CO3
3	A				
	1)	Select the correct option. 1. The expected number of trials needed to get the first success when p = 0.25 is: a) 0.25 b) 0.75 c) 4 d) 1 2.The variance of a Bernoulli random variable with success probability 0.6 is: a) 0.6 b) 0.36 c) 0.24 d) 0.4 3.The expected value of a discrete uniform random variable taking each integer value from 1 to 10 exactly once is: a) 5 b) 5.5 c) 6 d) 10 4.A bag contains 4 red and 6 black balls. Two balls are drawn one by one with replacement. The events “first ball red” and “second ball red”	5	L2	CO3

		are: a) Mutually exclusive b) Independent c) Dependent d) Complementary 5.From a well-shuffled deck of 52 cards, one card is drawn. The probability of drawing a red card is: a) 1/4 b) 1/2 c) 13/52 d) 26/52			
	2)	The probability function of a random variable X is given by $p(x) = k(4 - x)$ for $x = 1, 2, 3$ and $p(x) = 0$ elsewhere. Check whether p(x) is a legitimate pmf. If it is, find the value of k and $P(X \geq 2)$.	3	L2	CO3
	B				
	1)	Four weather stations A1, A2, A3, A4 predict rain with accuracies 85%, 80%, 90%, 75% and are consulted with equal probability. If rain occurs, determine the posterior probability that A3 predicted it correctly.	4	L3	CO2
	2)	Given $P(G) = 0.3$, $P(H\|G) = 0.9$, $P(H\|G') = 0.5$. Compute $P(G \cap H') + P(G' \mid H)$.	4	L3	CO1
	3)	A survey reveals that 55% of customers buy product P, 48% buy product Q, and 30% buy both P and Q. A customer is chosen at random from the survey. Calculate: (i) Probability that the customer buys at least one product ($P \cup Q$) (ii) Probability that the customer buys Q but not P (iii) Are the events "buying P" and "buying Q" independent? Give mathematical justification.	4	L3	CO1
SECTION II					
QN		Question	Marks	Level	CO
4	A				
	1)	Law of Large Numbers applies to: a) Any distribution with finite mean b) Only normal distribution c) Only discrete variables d) Only uniform	5	L2	CO5
	2)	Variance is always: a) Non-negative b) Negative c) Zero d) Greater than mean			CO4
	3)	For any random variable X, $F(\infty)$ equals: a) 0 b) 1 c) 0.5 d) ∞			CO3
	4)	For exponential distribution with mean 5, the parameter λ is: a) 5 b) 0.2 c) 1/5 d) 25			CO3
	5)	If $X \sim N(50, 16)$, then the standard deviation of X is: a) 4 b) 16 c) 50 d) 2			CO3
	6)	Check whether the function given below is a valid pdf: $f(x) = \begin{cases} 3/8\,(x^2 + 1) & \text{for } 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$ Perform the necessary integration and clearly state your conclusion with reason.	3	L2	CO3
	B				
	1)	The daily profit of a shop (in thousands) is normally distributed with mean 50 and variance 16. Find the approximate probability that the average profit over 64 days exceeds 52 thousand. Use CLT	4	L3	CO5
	2)	A bag contains 4 tickets numbered 1, 2, 3, 10. One ticket is drawn at random. Let X be the number on the ticket. (i) Write the pmf of X. (ii) Compute E(X) using the definition and also using the formula $E(X) = \Sigma\, x\, p(x)$. (iii) Find Var(X) and hence SD(X).	4	L3	CO4
	3)	The height of adult males in a country is normally distributed with mean 170 cm and standard deviation 8 cm. (i) Find the probability that a randomly selected male is taller than 180 cm. (ii) What height is exceeded by only 5% of males?	4	L3	CO3
5	A				

	1)	For independent continuous X and Y, E(X|Y=y) = a) E(X) b) E(Y) c) 0 d) Var(Y)	5	L2	CO6
	2)	The law of total probability for continuous RV is: a) $f_X(x) = \int f(x\|y)\, f_Y(y)\, dy$ b) $f_X(x) = \int f(x,y)\, dy$ c) Both a and b d) None			CO6
	3)	The law of total probability for discrete RV says: a) $p_X(x) = \Sigma_y\, p(x\|y)\, p_Y(y)$ b) $p_X(x) = \Sigma_y\, p(x,y)$ c) Both a and b d) None			CO6
	4)	For independent discrete X and Y: a) $p(x,y) = p_X(x) + p_Y(y)$ b) $p(x,y) = p_X(x) * p_Y(y)$ c) $p(x,y) = p_X(x) / p_Y(y)$ d) $p(x,y) = 1 - p_X(x)$			CO6
	5)	The marginal cdf of X is: a) $\int f(x,y)\, dy$ from $-\infty$ to ∞ b) $P(X \le x) = \int_{-\infty}^{x} \int_{-\infty}^{\infty} f(u,y)\, dy\, du$ c) $P(X > x)$ d) Sum over y of F(x,y)			CO6
	6)	State whether the following statements are TRUE or FALSE. 1. For discrete X and Y, P(x,y) can never be greater than 1. 2. The conditional pdf can be greater than the marginal pdf at the same point. 3. Conditional expectation E(X|Y) is itself a random variable.	3	L2	CO6 CO6 CO6
	B				
	1)	A triangular pdf is defined as f(x) = { 2 − 2x for $0 \le x \le 1$ { 0 otherwise Find the mean, $E(X^2)$, variance, and standard deviation of X.	4	L3	CO4
	2)	The joint probability mass function of X and Y is given by $p(x,y) = k(x^2 y)$ for x=1,2 and y=1,2,3 (a) Find the value of k. (1 mark) (b) Find the marginal pmfs of X and Y. (2 marks) (c) Find the conditional pmf of X given Y=2. (2 marks) (d) Compute $P(X \ge Y)$. (2 marks) (e) Are X and Y independent? Justify. (1 mark)	8	L3	CO6
6	A				
	1)	A sensor reading X is uniformly distributed on [5, 15]. What is the expected value E[X]? A) 5 B) 10 C) 15 D) 7.5	5	L2	CO4
	2)	A weather forecaster uses a joint distribution for X (Temperature, 10^0C or 20^0C) and Y (Humidity, 50\% or 80\%). The marginal probability P_Y(Y=80\%) = 0.50. The conditional probability P(X=20^0C | Y=80\%) = 0.6. What is the joint probability P(X=20^0C, Y=80\%)? A) 0.30 B) 0.10 C) 1.10 D) 0.40			CO6
	3)	The lifespan of a certain brand of battery is normally distributed with mean 100 hours and standard deviation 10 hours. Approximately, what percentage of batteries have a lifespan between 90 hours and 110 hours (i.e., within one standard deviation of the mean)? A) 50\% B) 68\% C) 95\% D) 99.7\%			CO3
	4)	The amount of time (in minutes) a video takes to buffer follows an Exponential distribution with mean 2 minutes. What is the expected variance of the buffering time? A) 2 minutes B) 4 minutes^2 C) 1 minute^2 D) 0.5 minutes^2			CO4
	5)	The daily number of complaints (X) at a cafe is P(X=0)=0.2, P(X=1)=0.5, P(X=2)=0.3. What is the expected number of complaints, E[X]? A) 1.0 B) 1.1 C) 0.8 D) 1.2			CO3
	6)	The battery life of a particular smartphone model is approximately normally distributed with a mean of 12 hours and a standard deviation of 1.5 hours. Three tested phones lasted: • Phone A: 14.25 hours • Phone B: 12 hours • Phone C: 9.75 hours	3	L2	CO3

		Calculate the Z-score for each phone's battery life.			
	B				
	1)	Calls arriving at a customer support line follow an Exponential Distribution. On average, a new call arrives every 2 minutes (i.e., the mean inter-arrival time is 2 minutes). What is the probability that the support line will have to wait more than 6 minutes before the next call arrives?	4	L3	CO3
	2)	Let X and Y be continuous random variables having joint pdf as follows: f(x,y) = { 10 ; 0 < x < 2, 2 < y < 3 { 0 ; otherwise Find the marginal pdf of X and of Y.	4	L3	CO6
	3)	The variables model two investment types: X (Return on Stock A, 0= Loss, 1= Gain) and Y (Return on Stock B, 0= Loss, 1= Gain). P(x,y) | Y=0 (Stock B Loss) | Y=1 (Stock B Gain) | PX(x) X=0 (Stock A Loss) | 0.45 | 0.15 | 0.6 X=1 (Stock A Gain) | 0.1 | 0.3 | 0.4 PY(y) | 0.55 | 0.45 | 1 Find the expected return on Stock B, given that there was a loss on Stock A. Find E[Y | X=0]	4	L3	CO6

SECTION III

QN		Question	Marks	Level	CO
7	A				
	1)	In cryptography, a birthday attack finds hash collisions in roughly: a) 2^n operations for n-bit hash b) 2^{n/2} operations c) n operations d) 2^{2n} operations	5	L2	CO7
	2)	A qubit state $\|\psi\rangle = \alpha\|0\rangle + \beta\|1\rangle$ is normalized if: a) $\|\alpha\| + \|\beta\| = 1$ b) $\|\alpha\|^2 + \|\beta\|^2 = 1$ c) $\alpha + \beta = 1$ d) $\alpha = \beta$			
	3)	PCA on covariance matrix is sensitive to: a) Units of measurement b) Only correlations c) Probability distributions d) Outliers only			
	4)	Predictive text suggests words with highest: a) Conditional probability b) Length c) Frequency only d) Alphabet order			
	5)	For 3D point cloud, PCA finds axes with variances 90, 8, 2 – reduce to 1D loses: a) All b) 10/100 = 10% variance c) 90% d) Nothing			
	6)	Identify the odd one out in each case. (i) a) Cryptographic hash b) Digital signature c) File deduplication d) Voice assistant prediction Odd one out: _____ Reason: _____ (ii) a) Markov assumption b) Limited context c) Sequence probability d) Dimensionality reduction Odd one out: _____ Reason: _____ (iii) a) Shor's algorithm b) Factoring c) Period finding d) Perplexity measure Odd one out: _____ Reason: _____	3	L2	CO7

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<tr><td></td><td>B</td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td>1)</td><td>Corpus:
1. I saw the big dog running.
2. I saw the small cat hiding.
3. She saw the huge mountain peak.
4. I saw the green grass grow.
5. He saw the big dog jump.
6. I saw the huge mountain top.
Question: Calculate the conditional probabilities and determine the most likely word to follow "I saw the".</td><td>4</td><td>L3</td><td>CO7</td></tr>
<tr><td></td><td>2)</td><td>Perform PCA on the given dataset. Determine the Principal Component 1 (PC1) and the percentage of variance it explains.
<table>
<tr><td>X1</td><td>X2</td></tr>
<tr><td>1</td><td>11</td></tr>
<tr><td>2</td><td>12</td></tr>
<tr><td>3</td><td>13</td></tr>
<tr><td>4</td><td>14</td></tr>
<tr><td>5</td><td>15</td></tr>
</table></td><td>8</td><td>L3</td><td>CO7</td></tr>
<tr><td>8</td><td>A</td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td>1)</td><td>If a hash table has 1,000,000 slots, collisions become likely (50% chance) after about how many entries?
a) 1,000,000 b) 500,000 c) Around 1,200 d) 100</td><td rowspan="5">5</td><td rowspan="5">L2</td><td rowspan="5">CO7</td></tr>
<tr><td></td><td>2)</td><td>Interference in quantum computing arises from:
a) Positive amplitudes only b) Amplitudes adding constructively or destructively c) Measurement always d) Classical probability</td></tr>
<tr><td></td><td>3)</td><td>First principal component captures:
a) Least variance b) Most variance c) Average variance d) No variance</td></tr>
<tr><td></td><td>4)</td><td>Smoothing in n-gram models handles:
a) Zero probabilities for unseen n-grams b) High frequencies only c) Long sequences d) Short words</td></tr>
<tr><td></td><td>5)</td><td>In 1,000 word corpus, "quick brown" 2 times, P(brown|quick) =:
a) 2/3 b) 1/1000 c) 0 d) 1</td></tr>
<tr><td></td><td>6)</td><td>Match the following real-world applications with their corresponding probability-based explanations. Each correct match carries 1 mark
Column A (Applications)
1. Factoring large numbers for encryption breaking
2. Clustering customer data in marketing
3. Autocomplete in email clients
Column B (Explanations) a. Periodic interference amplifies correct outcome probability b. Bigram models use immediate prior context for predictions c. Eigenvalues indicate proportion of total variability captured d. Birthday-like collisions in buckets despite uniform distribution e. Normalization ensures sum of squared amplitudes equals 1</td><td>3</td><td>L2</td><td>CO7</td></tr>
<tr><td></td><td>B</td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td>1)</td><td>Corpus:
1. The cat chased the mouse quickly.</td><td>4</td><td>L3</td><td>CO7</td></tr>
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2. The dog chased the ball hard.
3. The cat jumped on the fence.
4. The dog jumped over the gate.
5. The cat chased the tail fast.
Question: Using the Bigram model, calculate and compare the probability of the following two sentences. Which is more likely?
• Sentence A: "The cat chased the ball."
• Sentence B: "The dog jumped on."
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Perform PCA on the given dataset. Determine the Principal Component 1 (PC1) and the percentage of variance it explains.
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